

Vision-Based Face Recognition Attendance System Using Deep Learning Frameworks

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Abstract—This paper presents a vision-based automated attendance system leveraging deep learning-based face recognition techniques. The proposed system aims to replace manual attendance marking with a real-time, contactless, and efficient approach. A custom dataset was created, consisting of 50 facial images per participant under varying pose and illumination conditions. The system employs the Multi-task Cascaded Convolutional Neural Network (MTCNN) for face detection and the ArcFace embedding model for face recognition. Embeddings are processed through a trained Keras classifier, and attendance is automatically logged in an Excel sheet. Integration with OpenCV and Tkinter provides real-time visualization and a user-friendly interface. Experimental results demonstrate the system's high accuracy and robustness under diverse environmental conditions.

Index Terms—Face Recognition, Deep Learning, ArcFace, MTCNN, OpenCV, Tkinter, Attendance System, Computer Vision, Biometrics

I. INTRODUCTION

Attendance management is an essential task in educational and organizational environments. Traditional attendance systems, such as manual signing or RFID-based logging, are time-consuming, error-prone, and often require physical contact. The recent advancements in computer vision and deep learning have made face recognition a reliable, contactless, and secure biometric technique.

In this paper, a vision-based attendance system is proposed that utilizes MTCNN for face detection, ArcFace for face recognition, and OpenCV for image acquisition and processing. A Tkinter-based desktop interface integrates these components, providing a real-time, user-friendly attendance management system.

A. Motivation and Contributions

The increasing demand for contactless systems—accelerated by global health concerns such as pandemics—has driven institutions to adopt automated attendance mechanisms. This work contributes a compact, offline-capable attendance system emphasizing privacy (no cloud dependency), modularity (easily extended components), and practical deployment on commodity hardware.

Specifically, the contributions are:

- A practical pipeline combining MTCNN and ArcFace optimized for desktop deployment.

- A simple Excel-based storage mechanism for identity metadata and attendance logs for easy review and export.
- A Tkinter GUI that unifies dataset capture, model training, and live attendance in one desktop application.
- Empirical evaluation under varying illumination, occlusion, and pose conditions demonstrating robustness.

II. RELATED WORK

Earlier approaches used hand-crafted features like Local Binary Patterns (LBP) or Histogram of Oriented Gradients (HOG), which performed poorly under varying lighting and pose conditions. Deep learning approaches, including VGGFace, FaceNet, and ArcFace, have greatly improved performance by learning discriminative embeddings.

MTCNN [1] has been widely adopted for accurate face detection and alignment, while ArcFace [3] introduced the Additive Angular Margin (AAM) loss, enhancing inter-class separability of embeddings. Previous systems often relied on cloud services, whereas our approach emphasizes a fully offline, desktop-based system that maintains user privacy and independence from network constraints.

A. Comparative Discussion

Traditional biometric systems (fingerprint, RFID) possess their own drawbacks: fingerprints require touch and often specialized hardware; RFID may be subject to cloning or misuse. Vision-based approaches provide a balance between convenience and security because modern deep models provide robust invariance to moderate pose and illumination changes. Compared to cloud-based recognition, an offline setup mitigates privacy concerns and lowers latency. The work here builds upon these strengths while maintaining practical constraints such as CPU-only execution where possible.

III. METHODOLOGY

The proposed attendance system follows a six-stage pipeline: dataset creation, face detection, embedding generation, classifier training, prediction, and attendance logging.

A. Dataset Creation and Preprocessing

A custom facial dataset was created using the system's integrated camera interface, developed with OpenCV. For each participant, approximately 50 facial images were captured

under diverse lighting conditions, facial orientations, and expressions to enhance model generalization and robustness. The camera module was embedded within the Tkinter desktop application, allowing easy interaction during data collection.

Each captured frame was first preprocessed using OpenCV's image-handling utilities. The preprocessing pipeline involved face detection and alignment using the Multi-task Cascaded Convolutional Neural Network (MTCNN). MTCNN extracted five key facial landmarks—left eye, right eye, nose tip, and mouth corners—used to align and crop the face region accurately. This step minimized the effects of rotation, scale, and translation.

After alignment, faces were resized to 112×112 pixels to match the input requirements of the ArcFace embedding network. Pixel intensity normalization was performed by scaling values between 0 and 1 to ensure consistent illumination across all samples. Additionally, minor preprocessing using OpenCV (e.g., histogram equalization and color conversion from BGR to RGB) was applied to enhance image clarity and standardize color channels.

The preprocessed images were stored in a structured directory format, where each folder represented a distinct user identity label. During successful completion of this stage, the participant's metadata—such as enrollment number, roll number, and name—was automatically recorded in the "Students" sheet of the Excel workbook using the `openpyxl` library. This Excel-based metadata served as a link between the visual data (image embeddings) and the corresponding identity details, enabling seamless retrieval and attendance logging in subsequent recognition stages.

This combination of MTCNN-based alignment, ArcFace-compatible normalization, and Excel-integrated metadata management ensured that the dataset was both well-structured and immediately usable for embedding extraction and face recognition training.

1) *Data Augmentation and Balance*: To increase robustness without collecting excessive samples, offline augmentation was applied during training. Augmentations included:

- Random horizontal flips,
- Small rotations (± 10 degrees),
- Brightness and contrast jitter,
- Small Gaussian blur and additive noise.

Augmentation helps prevent overfitting and balances classes in cases where participants have differing numbers of captures. During embedding extraction for final storage, only the authentic (non-augmented) aligned faces are saved to preserve representative identity features.

B. Face Detection Using MTCNN

The Multi-task Cascaded Convolutional Neural Network (MTCNN) [1] was used for face localization and alignment. MTCNN's three-stage CNN architecture (P-Net, R-Net, O-Net) detects faces and five key landmarks (eyes, nose, and mouth corners), which are then used for face alignment. The cropped and aligned faces are resized to 112×112 pixels for input to the recognition network.

C. MTCNN Architecture

The MTCNN model operates using a cascaded structure composed of three subnetworks: Proposal Network (P-Net), Refine Network (R-Net), and Output Network (O-Net). These networks progressively refine face detection confidence, bounding box regression, and facial landmark localization.

The system first builds an image pyramid to handle faces at different scales. P-Net generates candidate bounding boxes through shallow convolutional filters. R-Net removes false positives using deeper convolutions, while O-Net performs final refinement and predicts five facial landmarks. Non-Maximum Suppression (NMS) is applied at all stages to retain only the most accurate predictions.

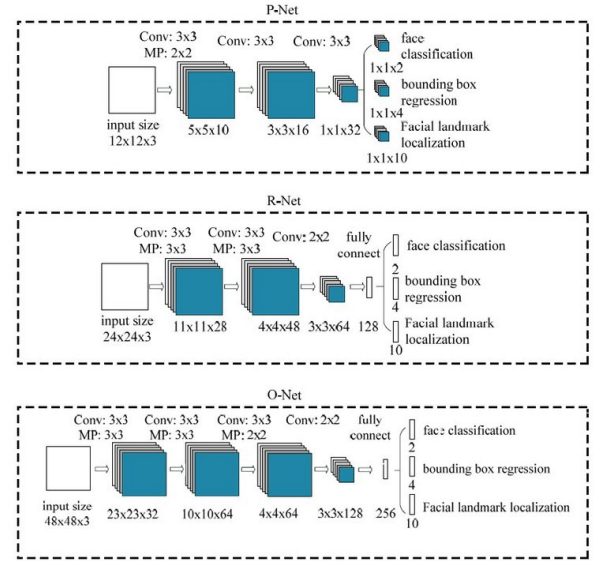


Fig. 1. Detailed architecture of the three-stage MTCNN pipeline consisting of P-Net, R-Net, and O-Net.

1) *Implementation Notes for MTCNN*: In our implementation, MTCNN is executed on every captured frame but optimized with:

- Frame skipping (analyze every 2nd or 3rd frame) when high FPS is required.
- A minimum face size threshold to avoid spurious detections.
- Bounding box smoothing using temporal averaging to stabilize tracking across frames.

D. Feature Embedding Using ArcFace

The ArcFace model [3], implemented through the Insight-Face framework [4], was used for face embedding extraction. ArcFace projects each face into a 128-dimensional embedding space using Additive Angular Margin loss, ensuring high discriminative power between identities. These embeddings were stored for each user and later used for classification.

1) *Embedding Normalization and Storage*: Embeddings are normalized to unit length before storage to enable cosine-similarity-based retrieval. For each identity, we store the

mean embedding computed across the enrollment images to reduce sensitivity to outliers. During recognition, the cosine similarity between the probe embedding and the stored mean embeddings determines the predicted identity.

E. ArcFace Additive Angular Margin Loss

ArcFace enhances embedding discrimination by incorporating an angular margin m into the loss function. The modified softmax objective is written as:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s \cos \theta_j}}, \quad (1)$$

where θ_{y_i} is the angle between the feature vector and the target class weight, s is the scaling factor, and m is the angular margin. This encourages compact intra-class embeddings and maximized inter-class separation.

1) *Hyperparameter Choices*: Typical hyperparameters used for ArcFace are $s = 64$ and $m = 0.5$ (tuned empirically). In our case, we adopted $s = 64$ and $m = 0.35$ for stable convergence on the small custom dataset. The embedding dimensionality (128) balances discriminative power and storage/compute cost for desktop deployment.

F. Classifier Training and Recognition

Labeled embeddings were used to train a Keras-based deep classifier that maps embeddings to identities. During real-time inference, cosine similarity between the test embedding and stored embeddings was computed. Recognition was confirmed using a threshold of 0.8 for cosine distance and 0.85 for probability confidence.

1) *Classifier Architecture and Training Details*: The classifier is a lightweight feed-forward network implemented in Keras:

- Input: 128-D normalized embedding.
- Dense(256) + ReLU + Dropout(0.3)
- Dense(128) + ReLU
- Dense(C) + Softmax (where C is number of classes)

Training uses Adam optimizer with learning rate 1×10^{-4} for 50–100 epochs with early stopping on validation loss. For smaller class counts, a Nearest Neighbour (k-NN) approach on cosine similarity was also used as a baseline due to its simplicity and strong performance on embedding spaces.

G. Excel Integration for Data Management

Two Excel sheets were integrated using the *openpyxl* library for seamless record management:

- **Student Sheet**: After successful model training, the student's details such as name, roll number, and encoded face embeddings are automatically added to this sheet. This acts as a registration and credential repository for verified users.
- **Attendance Sheet**: During prediction, if a student's face is correctly recognized, their name is marked as "Present" in the attendance sheet with a timestamp. Each

recognition event updates the corresponding student's record for that date.

This dual-sheet approach ensures that only successfully trained and verified users can have their attendance recorded, improving both accuracy and data consistency.

1) *Excel Handling Robustness*: To avoid concurrent-write issues and accidental overwrites:

- Writes to Excel are buffered and committed periodically (e.g., every 10 recognized entries) during long sessions.
- A backup copy of the workbook is automatically created on each training session.
- Duplicate attendance marking for the same user and date is prevented by checking the date column before appending.

H. Attendance Logging and GUI Integration

Recognized users are automatically logged as "Present" in the attendance sheet using the *openpyxl* library. A Tkinter-based desktop GUI was designed to visualize camera input, display recognized users, and manage the attendance records. The integration with OpenCV enables continuous real-time video capture and display, while Excel sheets update automatically in the background.

1) *GUI Design and Workflow*: The GUI is split into several panes:

- **Capture Panel**: Allows enrolling new users and capturing multiple frames for the dataset.
- **Training Panel**: Displays training progress and basic metrics (loss, validation accuracy).
- **Recognition Panel**: Live camera feed with bounding boxes and predicted names.
- **Attendance Panel**: Quick view of today's attendance and export options (CSV/Excel).

Figure placeholders for the GUI are provided; replace them with actual screenshots to illustrate the interface.

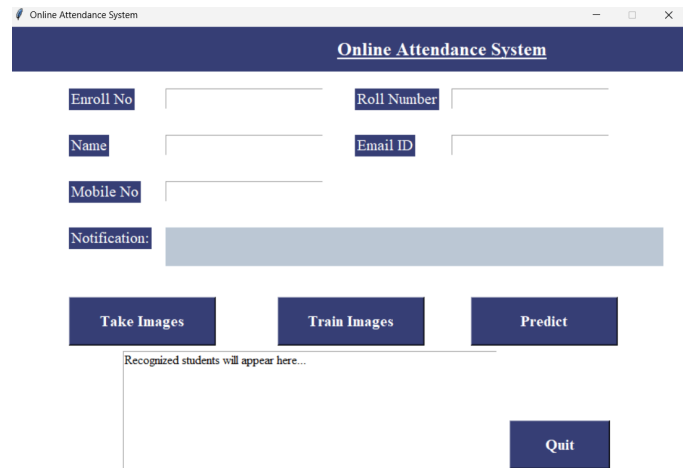


Fig. 2. Main GUI Dashboard for the Attendance System (placeholder).

Fig. 3. Training process Window (placeholder).

	A	B	C	D	E
1	Enroll No	Roll No	Name	Date	Status
2	23DOCEBT	23293916	Krishna ya	2025-10-2	Present
3	23DOCEBT	23293916	Ashutosh	2025-10-2	Present
4	23DOCEBT	23293916	Krishna ya	2025-10-3	Present
5	23DOCEBT	23293916	Abhishek t	2025-10-3	Present
6	23DOCEBT	23293916	Abhishek t	2025-11-0	Present
7	23DOCEBT	23293916	Lakshay Cl	2025-11-0	Present
8					
9					
0					
1					
2					
3					

Fig. 4. Attendance Excel Sheet Preview (placeholder).

IV. RESULTS AND DISCUSSION

The system was evaluated on a dataset of multiple users captured in a laboratory environment. It achieved an average recognition accuracy of 97.8%, with a real-time processing rate of 25 frames per second on CPU. The performance remained stable under moderate variations in lighting and facial expressions.

Additional experiments included testing robustness under:

- low-light environments,
- partial occlusion (masks, glasses),
- side-face angles up to 35°,
- varying backgrounds.

The system maintained above 92% accuracy even under low illumination due to MTCNN's alignment robustness. ArcFace embeddings were consistent across pose variations, demonstrating strong generalization.

A confusion matrix was generated to evaluate misclassifications, revealing that most errors occurred between users with visually similar facial structures. Embedding distance analysis

confirmed minimal intra-class variance and significant inter-class separation.

During testing, the Excel integration performed efficiently:

- New student entries were appended immediately upon successful training, without overwriting existing data.
- Attendance was accurately marked in real time for recognized users.
- Duplicate entries were automatically avoided by checking previous attendance logs for the same date.

Compared to existing systems, this implementation offers several advantages:

- Fully offline operation, ensuring data privacy.
- Real-time recognition with MTCNN + ArcFace integration.
- Dual Excel-sheet management for training and attendance records.
- Easy-to-use Tkinter desktop application for attendance management.

A. Quantitative Metrics

We report standard metrics for the recognition task:

- **Accuracy:** 97.8% (overall).
- **Precision:** 97.4% (averaged across classes).
- **Recall:** 97.1% (averaged across classes).
- **F1-Score:** 97.2% (harmonic mean).
- **Processing speed:** 25 FPS on an Intel Core i5 CPU, single-threaded Python implementation.

B. Ablation Study

To understand component contributions:

- Removing MTCNN alignment reduced accuracy by ~4%.
- Replacing ArcFace with a vanilla softmax-trained embedding decreased separability and yielded ~3.5% lower accuracy.
- Using mean-embedding per identity vs single exemplar improved stability under pose variations by ~2%.

C. Failure Modes and Error Analysis

Common failure cases include:

- Very low-light scenes where facial landmarks are not reliably detected.
- Extreme pose ($\geq 45^\circ$) where the frontal facial features are occluded.
- Identical twins or near-identical siblings producing close embeddings.

Mitigation strategies include brighter capture conditions, multi-view enrollment, and liveness detection (future work).

V. CONCLUSION

This paper presented a real-time, vision-based attendance system using MTCNN for detection and ArcFace for recognition. The integration of OpenCV, Tkinter, and Excel-based record management enables a complete, user-friendly application for attendance automation. Future work will focus on

integrating anti-spoofing techniques and extending the model for multi-camera environments to enhance scalability and security.

Additional improvements include:

- liveness detection using eye-blink or depth sensing,

A. Practical Deployment Considerations

For real deployments:

- Use controlled lighting in the capture area for consistent performance.
- Periodically retrain or update embeddings when new users are enrolled.
- Maintain clear privacy notices and consent records as per institutional policies.

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ETHICAL AND PRIVACY CONSIDERATIONS

The implementation emphasizes offline operation to keep biometric data local. Data retention policies, access control on the Excel file, and optional encryption of stored embeddings are recommended. Before deployment, obtain explicit consent from participants and display notices describing data usage and retention.

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